Finding Waldo with ML

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Contents

- Objective
- Data & Preprocessing
- Training
- Output & Performance

Objective

Q: *What is the function we want our model to have?*

Take an input image, look for **Waldo** and potentially for **lookalikes we are supposed to search for** (such as **Wenda** or **Wilma**), mark them on the image

Objective

Q: *How is ‘correctness’ judged in the context of this task?*

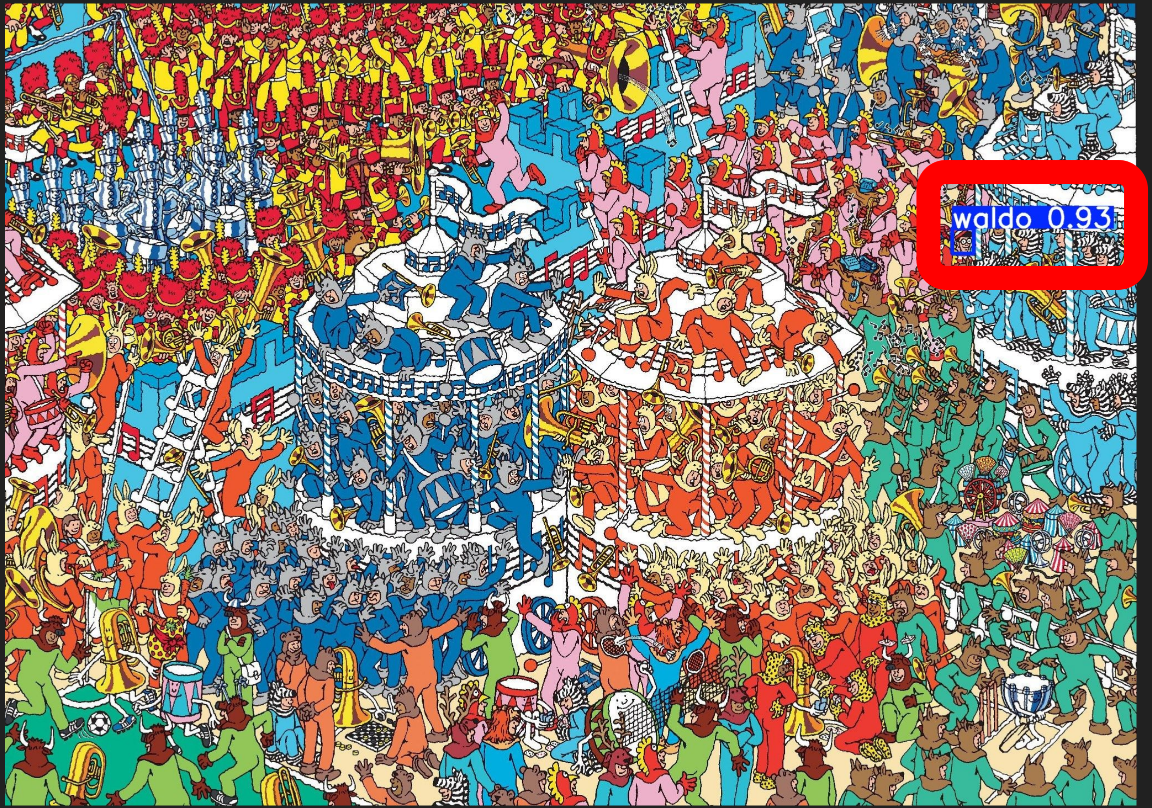
Output is **correct** if it:

- ❖ **Labels true Waldo** with confidence ≥ 80%
- ❖ Does not have large amount of wrong labels (labels something which is neither Waldo or his lookalikes)

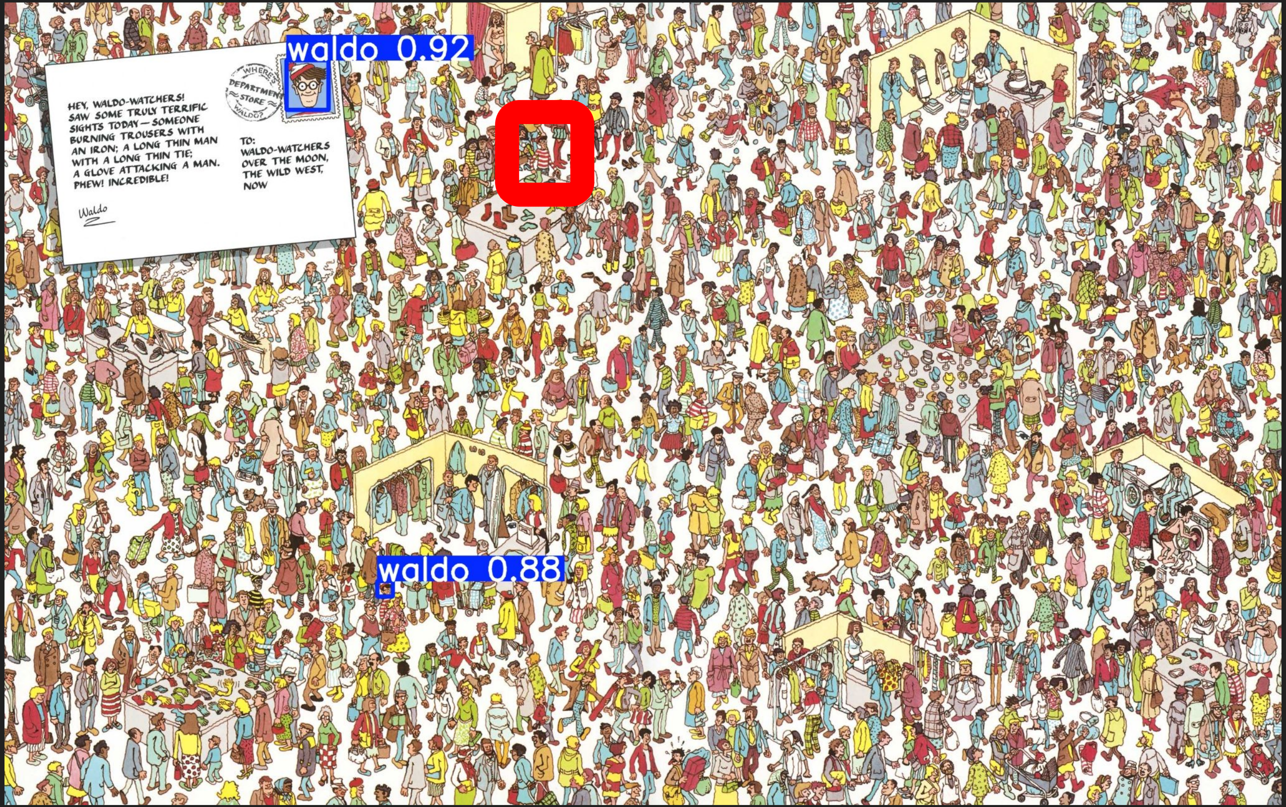
Output is **partially correct** if it:

- ❖ **Labels Waldo lookalikes** with confidence ≥ 80%
- ❖ Does not have …

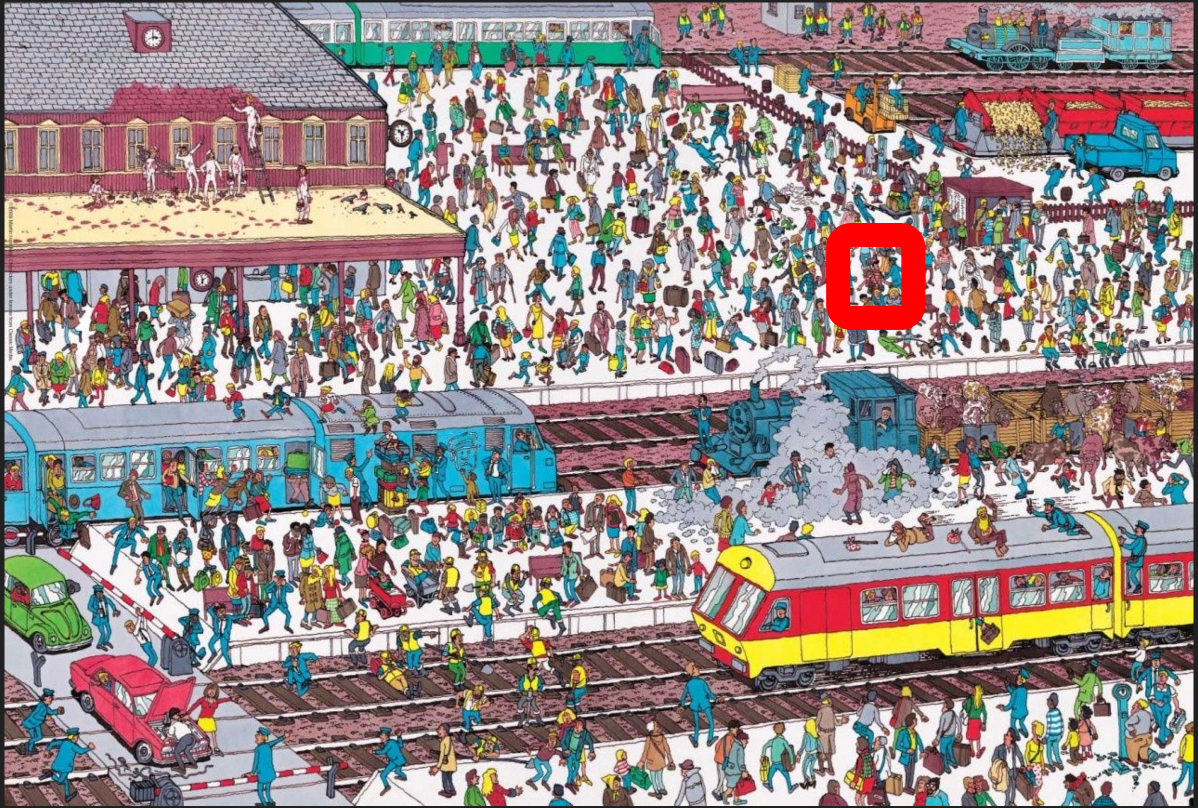
Correct Output



Partially Correct Output



Incorrect Output

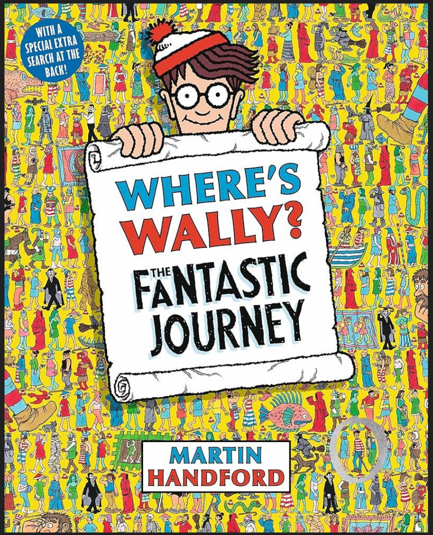


Data & Preprocessing

Input dataset:

- ❖ Where’s Wally; The Fantastic Journey (edition published in 2007)
- ❖ Subset of Images containing waldo from [github]

In total less than 45 images … too small



Data & Preprocessing

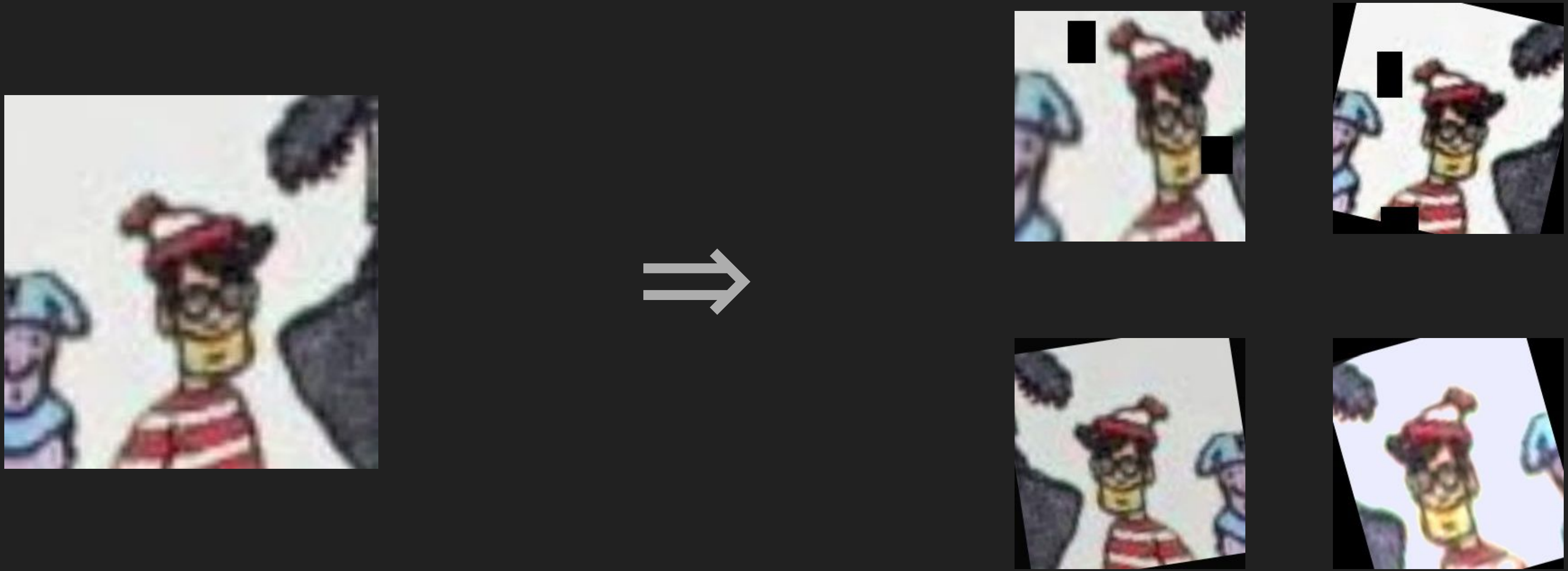
Preprocessing & Augmentations

- ❖ Cropped images to be 256x256 (`src/crop-images.py`)
- ❖ Added flipped variants of some images (different head orientation)
- ❖ Added rotated variants of some images (different head tilt)
- ❖ Added blur, saturation & perspective shift, ‘holes’ in the image (different page scan qualities, Waldo’s head being partially covered, …)

  (`src/augment-waldo.py`)

⇒ more images to work with

Data & Preprocessing



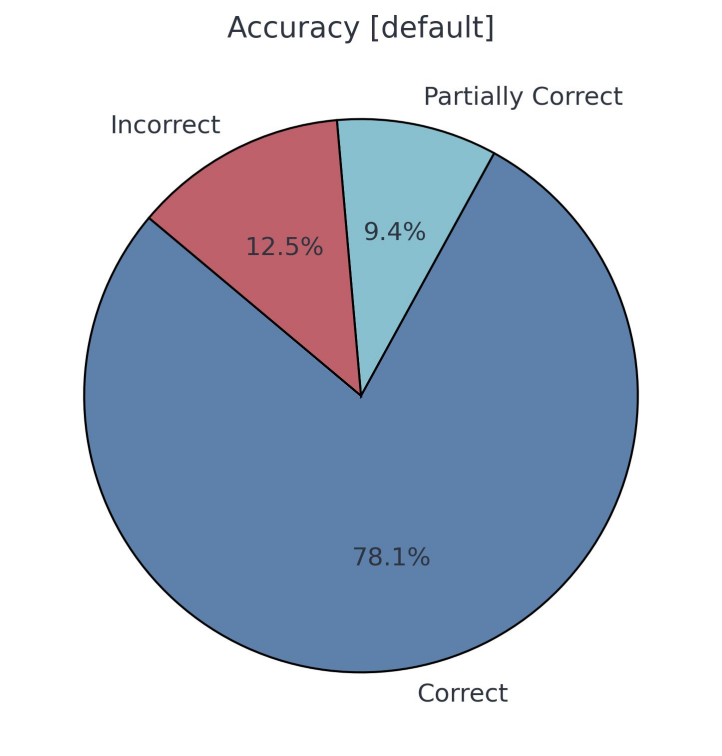
Training

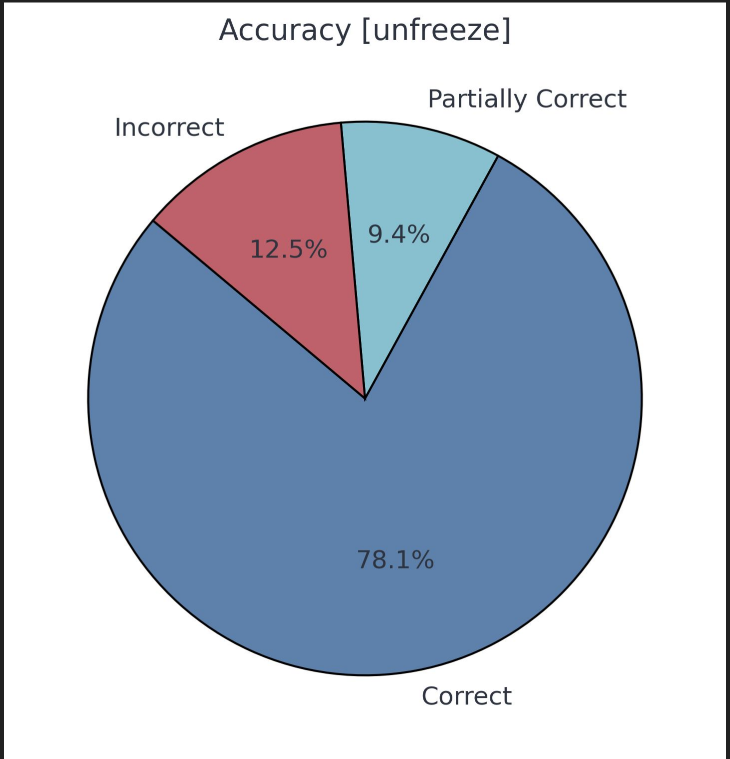
Used **YOLOv8** as a base for **transferred learning** with waldo dataset

Different approaches tried, namely:

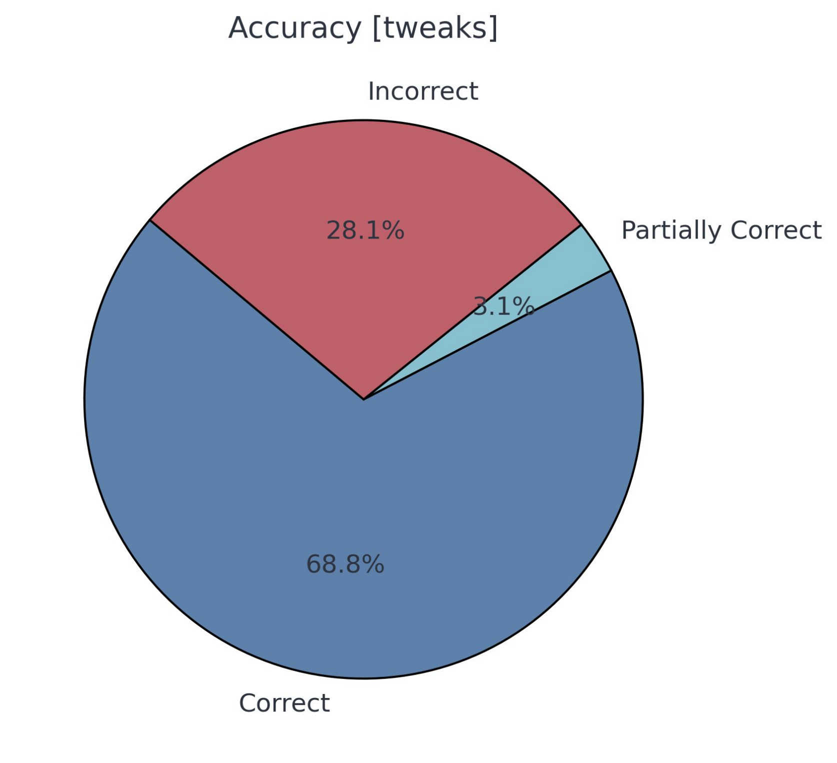
- ❖ Default – Unfreezing only ‘head’ layers ⇒ keeps feature extraction capabilities of YOLO model, but adapts it to this task’s use case
- ❖ Unfreezing all layers
- ❖ Only tweaking training params & optimizer (adam)
- ❖ Using build-in tune function (SLOW)

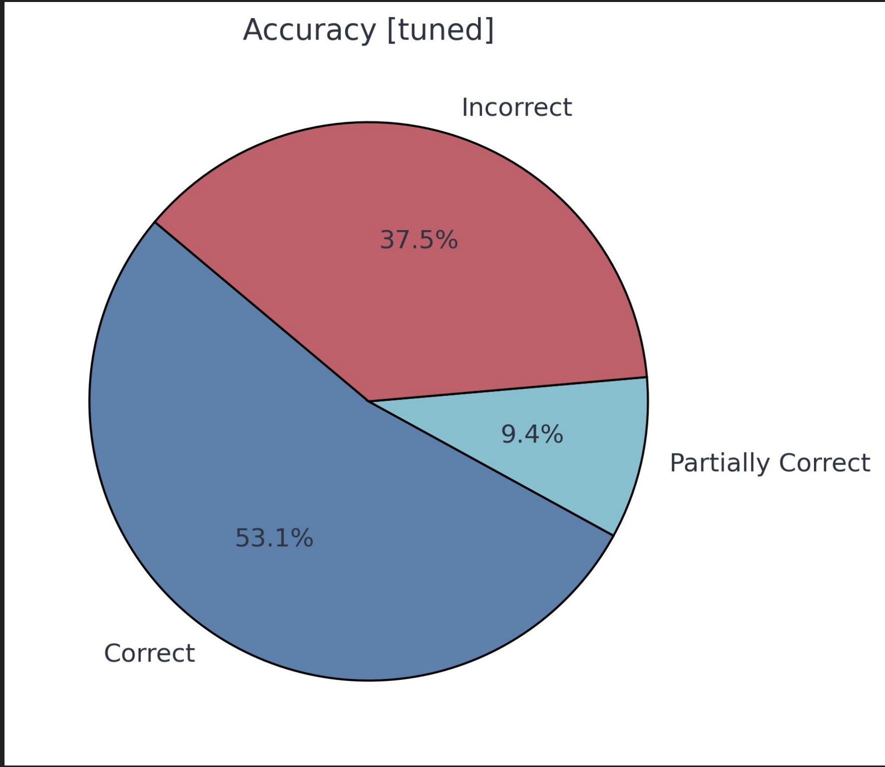
Output & Performance





Output & Performance





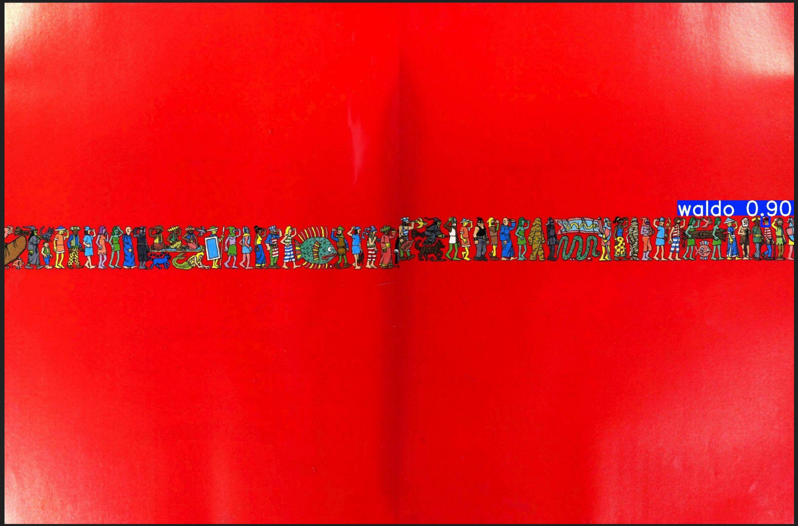
Output & Performance

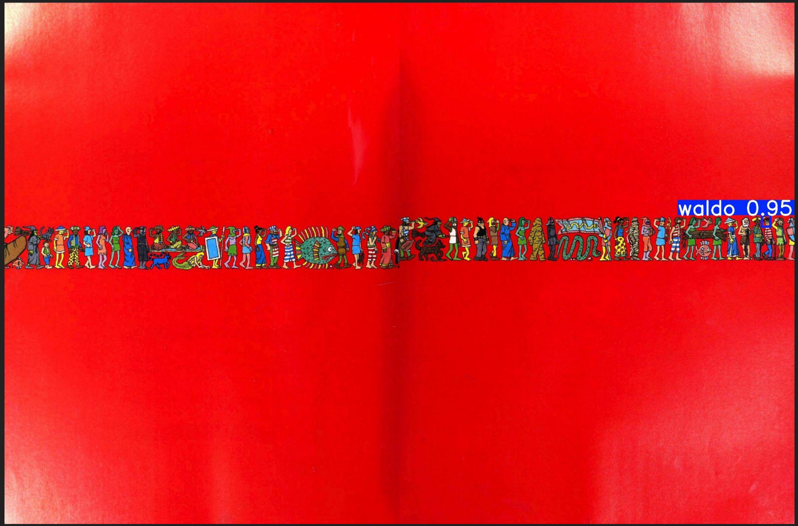
Main Takeaways:

- ❖ Training with default configuration led to more consistent results
- ❖ Vast majority of wrong guesses were when not detecting anything, rather than detecting a wrong thing
- ❖ While not as consistent, the tweaked/tuned models were more confident in detecting waldo, when they detected him

Output & Performance

Output & Performance





Conclusion

Given the circumstances – small dataset, limited time for training – I'd consider the final output models to be **successful in finding**, or at the very least **providing pointers towards probable waldo characters**.

Possible improvements

Gathering much more data (via book scans if possible) and/or creating synthetic generated data

More 'educated guesswork' 🙂 when tweaking training parameters

Thank you for your attention